

Extra Credit Assignment 1

Virtualization and Containers

Instructions:

- Assigned date: Thursday February 22nd 2024
- Due date: 11:59pm on Thursday March 7th 2024
- Maximum points: 200
- This programming assignment must be done individually.
- Please post your questions on the D2L Programming Assignments topic.
- Only the report must be uploaded to D2L.
- Late submission will be penalized at 20% per late day.

This programming assignment aims to make you more familiar with virtualization and container technologies. This assignment is split into two parts. In the first part you need to configure 5 LXD/Docker containers on a Ubuntu 22.04 server machine and in the second part you need to configure 5 LXD/KVM/VirtualBox virtual machines on the same machine. The Ubuntu machine that will host the container and the virtual machines can be either an instance from Chameleon Cloud or a Virtual Machine hosted on your personal computer with at least 4 cores, 10GiB of RAM and at least 400 GB of storage. **You need to complete all of the configurations from the command line!**

1 Containers (50%)

In part one of the assignment you need to configure 5 containers using either the LXD or the Docker container technologies. The containers need to be named: **client1**, **client2**, **client3**, **client4** and **server**. The **server** container needs to be configured with 4 cores and 2GiB of RAM and may be connected to the default bridge/NAT network. The **client** containers need to be configured with 2 cores and 2GiB of RAM each and they may be connected to the default bridge/NAT network. You have to make sure that all of the containers are connected to the same network. All of the containers must run Ubuntu 22.04 server edition.

After you configured the containers, distributed the data from the 5 datasets that you've been using for the programming assignments in the following way:

- Dataset1 needs be copied to **client1**.
- Dataset2 needs to be copied to **client1** and **client2**, where each client receives half of the data.

- Dataset3 needs to be copied to **client1** and **client2**, where each client receives half of the data.
- Dataset4 needs to be copied to **client1**, **client2**, **client3** and **client4**, where each client receives a quarter of the data.
- Dataset5 needs to be copied to **client1**, **client2**, **client3** and **client4**, where each client receives a quarter of the data.

OPTIONAL: Clone your code from programming assignment 3 to all containers and run the following experiments:

- Start the server program on the **server** container. Start the client program on the **client1** container and connect to the server. Index all of the folders from Dataset1 and perform one search operation with one term and one search operation with two terms (i.e. an AND search query).
- Start the server program on the **server** container. Start the client programs on the **client1** and **client2** containers and connect them to the server. Index all of the folders from Dataset2 from both clients and perform one search operation with one term and one search operation with two terms (i.e. an AND search query) from only one client.
- Start the server program on the **server** container. Start the client programs on the **client1** and **client2** containers and connect them to the server. Index all of the folders from Dataset3 from both clients and perform one search operation with one term and one search operation with two terms (i.e. an AND search query) from only one client.
- Start the server program on the **server** container. Start the client programs on the **client1**, **client2**, **client3** and **client4** containers and connect them to the server. Index all of the folders from Dataset4 from all clients and perform one search operation with one term and one search operation with two terms (i.e. an AND search query) from only one client.
- Start the server program on the **server** container. Start the client programs on the **client1**, **client2**, **client3** and **client4** containers and connect them to the server. Index all of the folders from Dataset4 from all clients and perform one search operation with one term and one search operation with two terms (i.e. an AND search query) from only one client.

Answer the following questions and provide screenshots of the commands that you used to configure the containers:

- Which container technology did you use and how did you install it on the Ubuntu machine?
- How did you create the 5 containers and make sure that they had the proper amount of cores and RAM?
- How did you make sure that the containers are connected to the same network?
- How did you copy the datasets to the clients?
- **OPTIONAL:** How did you run the programs on the containers?

- What was the most challenging aspect of configuring containers?

2 Virtual Machines (50%)

In part two of the assignment you need to configure 5 virtual machines using either the LXD or the QEMU/KVM or the VirtualBox virtualization technologies. The virtual machines need to be named: **client1**, **client2**, **client3**, **client4** and **server**. The **server** virtual machine needs to be configured with 4 cores, 2GiB of RAM and 60GB of storage and may be connected to the default bridge/NAT network. The **client** virtual machines need to be configured with 2 cores, 2GiB of RAM and 60GB of storage each and they may be connected to the default bridge/NAT network. You have to make sure that all of the virtual machines are connected to the same network. All of the virtual machines must run Ubuntu 22.04 server edition.

After you configured the virtual machines, distributed the data from the 5 datasets that you've been using for the programming assignments in the following way:

- Dataset1 needs to be copied to **client1**.
- Dataset2 needs to be copied to **client1** and **client2**, where each client receives half of the data.
- Dataset3 needs to be copied to **client1** and **client2**, where each client receives half of the data.
- Dataset4 needs to be copied to **client1**, **client2**, **client3** and **client4**, where each client receives a quarter of the data.
- Dataset5 needs to be copied to **client1**, **client2**, **client3** and **client4**, where each client receives a quarter of the data.

OPTIONAL: Clone your code from programming assignment 3 to all virtual machines and run the following experiments:

- Start the server program on the **server** virtual machines. Start the client program on the **client1** virtual machine and connect to the server. Index all of the folders from Dataset1 and perform one search operation with one term and one search operation with two terms (i.e. an AND search query).
- Start the server program on the **server** virtual machine. Start the client programs on the **client1** and **client2** virtual machines and connect them to the server. Index all of the folders from Dataset2 from both clients and perform one search operation with one term and one search operation with two terms (i.e. an AND search query) from only one client.
- Start the server program on the **server** virtual machine. Start the client programs on the **client1** and **client2** virtual machines and connect them to the server. Index all of the folders from Dataset3 from both clients and perform one search operation with one term and one search operation with two terms (i.e. an AND search query) from only one client.

- Start the server program on the **server** virtual machine. Start the client programs on the **client1**, **client2**, **client3** and **client4** virtual machines and connect them to the server. Index all of the folders from Dataset4 from all clients and perform one search operation with one term and one search operation with two terms (i.e. an AND search query) from only one client.
- Start the server program on the **server** virtual machine. Start the client programs on the **client1**, **client2**, **client3** and **client4** virtual machine and connect them to the server. Index all of the folders from Dataset4 from all clients and perform one search operation with one term and one search operation with two terms (i.e. an AND search query) from only one client.

Answer the following questions and provide screenshots of the commands that you used to configure the virtual machines:

- Which virtual machine technology did you use and how did you install it on the Ubuntu machine?
- How did you create the 5 virtual machines and made sure that they had the proper amount of cores and RAM?
- How did you make sure that the virtual machines are connected to the same network?
- How did you copy the datasets to the clients?
- **OPTIONAL:** How did you run the programs on the virtual machines?
- What was the most challenging aspect of configuring virtual machines?

3 Datasets

You need to copy the [Gutenberg Project Datasets](#) to the clients as part of this assignment, after you create and configure the containers and the virtual machines. There are 5 datasets with increasing size and number of files. Each dataset contains a number of 16 folder each containing a collection of ASCII TXT documents representing books collected from the Internet.

4 Chameleon Cloud or Virtual Machine Testbed

You may use use a Virtual Machine running Ubuntu 22.04 with at least 4 cores, 10GiB of RAM and 400GB of storage, or you may use the Chameleon Testbed to set up the containers and the virtual machines. The Chameleon instance needs to be of one of these types:

- compute.icelake_r650 or compute.icelake_r750
- compute.cascadelake or compute.cascadelake_r
- compute.skylake
- compute.haswell

5 Where you will submit

You will have to submit your report to D2L.

6 What you will submit

As you work on the configuration aspect of the container and of the virtual machines you should take screenshots of the commands that you used. When you have finished configuring the container and the virtual machines, answer the questions from each part and include the screenshots in the answers. Include all of the answers and the screenshots in the report. You need to hand in:

1. **Report (100%):** A written PDF document describing the configurations of the containers and the virtual machines, along with screenshots and text answers to the assignment questions. Make sure to label the answers with the appropriate part and question number.

Do not upload the datasets!

Submit step 1: report through D2L.

Grades for late submission will be lowered 20% per late day.